Advantages of natural vegetative reproduction

- Only one parent is required for propagation.
- The new individual plants produced are genetically identical.
- In some plants, this enables to spread rapidly. Example: *Spinifex*
- Horticulturists and farmers utilize these organs of natural vegetative reproduction for cultivation and to harvest plants in large scale.

Disadvantage of natural vegetative reproduction

- New plants produced have no genetic variation.

1.2.2 Artificial Methods

Apart from the above mentioned natural methods of vegetative reproduction, a number of methods are used in agriculture and horticulture to propagate plants from their parts. Such methods are said to be artificial propagation. Some of the artificial propagation methods have been used by man for a long time and are called **conventional methods**. Now-a-days, technology is being used for propagation to produce large number of plants in a short period of time. Such methods are said to be **modern methods**.

A. Conventional methods

The common methods of conventional propagation are cutting, grafting and layering.

a. Cutting: It is the method of producing a new plant by cutting the plant parts such as root, stem and leaf from the parent plant. The cut part is placed in a suitable medium for growth. It produces root and grows into a new plant. Depending upon the part used it is called as root cutting (*Malus*), stem cutting (*Hibiscus, Bougainvillea* and *Moringa*) and leaf cutting (*Begonia, Bryophyllum*). Stem cutting is widely used for propagation.

b.Grafting: In this, parts of two different plants are joined so that they continue to grow as one plant. Of the two plants, the plant which is in contact with the soil is called **stock** and the plant used for grafting is called **scion** (Figure 1.2 a). Examples are Citrus, Mango and Apple. There are different types of grafting based on the method of uniting the scion and stock. They are bud grafting, approach grafting, tongue grafting, crown grafting and wedge grafting.

i. Bud grafting: A T- shaped incision is made in the stock and the bark is lifted. The scion bud with little wood is placed in the incision beneath the bark and properly bandaged with a tape.

ii. Approach grafting: In this method both the scion and stock remain rooted. The stock is grown in a pot and it is brought close to the scion. Both of them should have the same thickness. A small slice is cut from both and the cut surfaces are brought near and tied together and held by a tape. After 1-4 weeks the tip of the stock and base of the scion are cut off and detached and grown in a separate pot.

iii. Tongue grafting

A scion and stock having the same thickness is cut obliquely and the scion is fit into the stock and bound with a tape.

iv. Crown grafting.

When the stock is large in size scions are cut into wedge shape and are inserted on the slits or clefts of the stock and fixed in position using graft wax.

v. Wedge grafting

In this method a slit is made in the stock or the bark is cut. A twig of scion is inserted and tightly bound so that the cambium of the two is joined.

Activity

Visit a nursery, observe the method of grafting, layering and do these techniques with plants growing in your school or home

c. Layering: In this method, the stem of a parent plant is allowed to develop roots while still intact. When the root develops,

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